

Interacting Multipoles of Rank 1 and 4

$$\begin{aligned}
 & + \frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_{\alpha}^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot \\
 & \left[\begin{aligned}
 & -945 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A
 \end{aligned} \right] \tag{1}
 \end{aligned}$$

Simplify deltas:

$$\begin{aligned}
& + \frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_{\alpha}^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{aligned}
& -945 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \delta(\delta, \delta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\delta} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\epsilon} \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\gamma\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\alpha} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\gamma} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\beta} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_{\gamma} \cdot R_{\delta} \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\gamma\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\gamma\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\delta} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_{\epsilon} \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A
\end{aligned} \right] \quad (2a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{aligned}
& -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\epsilon}^A
\end{aligned} \right] \quad (2b)
\end{aligned}$$

$$\begin{aligned} & \left[\begin{aligned} & -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ & +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \\ & -15 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \end{aligned} \right] \end{aligned} \quad \xrightarrow{\text{reduced indices}} R_z^{-11} \cdot \frac{1}{105} . \quad (2c)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{aligned} &-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\ &+210 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\ &+315 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ &+420 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\ &-45 \cdot R_z^4 \cdot R_\alpha \cdot \mu_\alpha^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ &-45 \cdot R_z^4 \cdot R_\beta \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \\ &-135 \cdot R_z^4 \cdot R_\gamma \cdot \mu_\alpha^B \cdot \Phi_{\alpha\beta\beta\gamma}^A \end{aligned} \right] \quad (2d)$$

Simplify R:

$$+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_{\alpha}^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{aligned} &-945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zzzz}^A \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\delta\delta}^A \\ &+210 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\gamma\gamma}^A \\ &+315 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \Phi_{zz\beta\beta}^A \\ &+420 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_{\alpha}^B \cdot \Phi_{zzz\alpha}^A \\ &-45 \cdot R_z^4 \cdot R_z \cdot \mu_z^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ &-45 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\ &-135 \cdot R_z^4 \cdot R_z \cdot \mu_{\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \end{aligned} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot \begin{bmatrix} +420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 45 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\ -135 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A \end{bmatrix} \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-11} \cdot \frac{1}{105} \cdot \left[\begin{aligned} &+420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 45 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A \\ &-135 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A \end{aligned} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot \frac{1}{105} \cdot [+420 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{zzz\alpha}^A - 180 \cdot R_z^5 \cdot \mu_\alpha^B \cdot \Phi_{z\alpha\beta\beta}^A] \quad (3d)$$

Simplify Dipoles:

$$+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_\alpha^B \Phi_{\beta\gamma\delta\epsilon}^A = R_z^{-11} \cdot \frac{1}{105} \cdot [+420 \cdot R_z^5 \cdot \mu_y^B \cdot \Phi_{yzzz}^A - 180 \cdot R_z^5 \cdot \mu_y^B \cdot \Phi_{yz\beta\beta}^A] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot \frac{1}{105} \cdot [+0] \quad (4b)$$

Combining everything:

$$+\frac{1}{105} \cdot T_{\alpha\beta\gamma\delta\epsilon} \mu_\alpha^B \Phi_{\beta\gamma\delta\epsilon}^A = +0 \quad (5)$$